

DATA 630: Scalable Machine Learning Systems & Cloud AI Architectures

School of Computing & Data Engineering • Fall Term 2026

LEAD INSTRUCTOR

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COURSE STRUCTURE

Credits: 4.0
Grading: Letter / Percentage (0–100%) Turing Lab 304 & Virtual Stream

OFFICE & LAB HOURS

Thursdays 10:00 AM – 12:30 PM PST
Turing Lab 304 & Virtual Stream

COURSE CATALOG DESCRIPTION

This graduate-level systems course provides an in-depth study of distributed machine learning pipelines, scalable model training, inference serving, and enterprise MLOps architectures. Students will implement real-time feature stores, containerized model deployment with Kubernetes, distributed gradient descent using PyTorch DDP and Ray, and automated continuous integration/continuous deployment (CI/CD) pipelines for foundation models.

PROGRAM LEARNING OUTCOMES & TECHNICAL COMPETENCIES

- **PLO 1 (Distributed Infrastructure):** Architect and scale distributed model training clusters across multi-node GPU instances using PyTorch DDP and Ray Train.
- **PLO 2 (Production Serving & Low Latency):** Deploy high-throughput, sub-100ms model inference endpoints utilizing Triton Inference Server, vLLM, and Redis cache clusters.
- **PLO 3 (Continuous Pipeline Orchestration):** Implement reproducible CI/CD for machine learning (MLOps) with automated data drift monitoring and model registry management.
- **PLO 4 (Ethical AI & Data Governance):** Implement audit logging, data lineage tracking, privacy-preserving machine learning, and cost-optimization governance in cloud environments.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

ASSIGNMENT DESCRIPTION	WEIGHT	DUE MODULE	DELIVERABLE FORMAT
Distributed Training Benchmark Project	30%	Module 05	GitHub Repo, Profiling Logs & 8-Page Whitepaper
Production Inference Microservice & Load Test	25%	Module 08	Containerized Service, Benchmark Dashboard & Dockerfile
Automated End-to-End MLOps Pipeline Capstone	30%	Module 11	Live Production Deployment, IaC Manifests & Final Pitch
Architecture Seminar & Code Reviews	15%	Continuous	Bi-Weekly Lab Code Reviews & Tech Briefs
Total Course Assessment	100%	Term End	

ASSIGNMENT DESCRIPTION	WEIGHT	DUE MODULE	DELIVERABLE FORMAT
			Comprehensive Cumulative Evaluation

ASSIGNMENT SPECIFICATIONS

1. Distributed Training Benchmark Project (30%)

Students implement and profile data-parallel and pipeline-parallel distributed training routines for a 3-billion-parameter Transformer architecture using PyTorch DDP and Ray on a multi-GPU cluster. The deliverable includes a reproducible GitHub repository, distributed GPU utilization flame graphs, scaling efficiency analysis, and an 8-page IEEE-format technical whitepaper.

2. Production Inference Microservice & Load Test (25%)

Students containerize a quantized deep learning model and expose low-latency gRPC/REST endpoints using Triton Inference Server or vLLM. The deliverable requires Locust or k6 load test results demonstrating sustained throughput under 1,000 concurrent RPS with p99 latency < 85ms, complete with Prometheus/Grafana monitoring configurations.

3. Automated End-to-End MLOps Pipeline Capstone (30%)

In teams of two, students design a complete cloud-native pipeline: automated feature engineering (Feast), model tracking and registry (MLflow), scheduled continuous training (Kubeflow/Airflow), automated Canary deployment on Kubernetes, and drift detection (Evidently AI). Includes live deployment validation and an oral technical pitch.

4. Architecture Seminar & Code Reviews (15%)

Evaluated through structured peer code reviews in GitHub pull requests, participation in technical whitepaper discussions, and bi-weekly architecture decision records (ADRs) submitted during lab sessions.

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

TIMELINE	MODULE	TECHNICAL TOPIC FOCUS	ASSIGNED READINGS & DOCS
Week 01	Module 01	ML Systems Architecture & Data Ingestion Pipelines	Huyen (Ch. 1–3); Kleppmann (Ch. 1)
Week 02	Module 02	Feature Engineering at Scale & Feature Stores	Huyen (Ch. 4 & 5); Feast Architecture Specs
Week 03	Module 03	Distributed Data-Parallel Training (DDP & Horovod)	PyTorch Distributed Docs; Li et al. (2020)
Week 04	Module 04	Model Parallelism, ZeRO & DeepSpeed Optimization	Rajbhandari et al. (2020); Shoeybi et al. (Megatron)
Week 05	Module 05	Model Compression: Quantization, LoRA & Distillation	Dettmers et al. (QLoRA); Hu et al. (LoRA)
Week 06	Module 06	Serving Infrastructure & Continuous Batching	vLLM Technical Paper; Triton User Guide

TIMELINE	MODULE	TECHNICAL TOPIC FOCUS	ASSIGNED READINGS & DOCS
Week 07	Module 07	Container Orchestration & Kubernetes for Machine Learning	Burns et al. (Ch. 4–6); Kubeflow Operator Docs
Week 08	Module 08	Monitoring, Observability & Data Drift Detection	Huyen (Ch. 8 & 9); Evidently AI Whitepaper
Week 09	Module 09	Workflow Orchestration with Airflow & Prefect	Huyen (Ch. 6); Apache Airflow Core Architecture
Week 10	Module 10	Cost Governance, Spot Instances & Cloud FinOps for AI	Cloud AI Pricing Guides; Stoica et al. (Ray Paper)
Week 11	Module 11	Final Capstone Showcase, Live Demos & Term Retrospective	Industry System Architecture Whitepapers

Late Submission & Extension Policy: Production releases wait for no one. Late code submissions incur an automatic 5% deduction per 24 hours late (up to 72 hours max). Extension requests must be submitted via formal instructor communication at least 48 hours in advance of the deadline.

COURSEPAL PARSER & APPLICATION MAPPING GUIDE

This document structure has been optimized for clean automated parsing into your CoursePal architecture. Below are the precise app views and fields populated upon upload:

1. ASSIGNMENTS & GRADES DASHBOARD

- **Distributed Training Benchmark:** Maps as a major technical deliverable, weight: 30%, due Module 05.
- **Production Inference Microservice:** Maps as a systems lab benchmark, weight: 25%, due Module 08.
- **MLOps Pipeline Capstone:** Maps as the term capstone project, weight: 30%, due Module 11.
- **Seminar & Code Reviews:** Maps as a continuous recurring contribution, weight: 15%.

2. WEEKLY READING & RESOURCE TRACKER

- Extracts 11 weekly reading blocks into clickable to-do checkboxes (e.g., *Huyen Ch. 1–3*, *PyTorch Distributed Docs*, *vLLM Paper*).
- Automatically indexes engineering docs and architectural whitepapers into the course resource drawer.

3. SMART CALENDAR & REMINDER ALERTS

- Schedules 11 weekly Thursday lecture and lab sessions (10:00 AM – 12:30 PM PST).
- Configures automated 48-hour deadline alert triggers matching the course extension window.